

# David Silva

## SENIOR SOFTWARE ENGINEER

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## PROFESSIONAL SUMMARY

Senior software engineer with 15+ years building scalable, distributed systems serving millions of users across government, fintech, and high-growth startups. Deep expertise in Ruby and Rails, event-driven architecture, AI integration, and platform reliability at scale. Currently working across the architecture, observability, AI, and reliability of an FCA-regulated mortgage platform processing 1.7M requests a day. I build systems that **earn their complexity through evidence, not fashion**. Proven track record of measurable impact, including an **80% reduction in support tickets** and customer-facing AI on platforms serving **1M+ users**.

## KEY SKILLS

|                        |                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| LANGUAGES & FRAMEWORKS | Ruby/Rails, JavaScript, SQL, Python                                                                         |
| ARCHITECTURE           | Distributed Systems, Event-Driven Architecture, Modular Monoliths / Rails Engines, Microservices, REST APIs |
| DATA & MESSAGING       | PostgreSQL, MySQL, Redis, Cassandra, Sidekiq, Kafka, Elasticsearch, Snowflake                               |
| CLOUD & AI             | AWS (EC2, S3, Lambda, RDS), AWS Bedrock, LLM integration                                                    |
| OBSERVABILITY & DEVOPS | NewRelic, Datadog, Sentry, Docker, GitHub Actions, CircleCI                                                 |
| PRACTICES              | TDD, Pair Programming, Agile/Scrum, Technical Mentoring, Onboarding                                         |

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer, Tembo** Feb 2026 – Present

Working across the architecture, observability, AI, and reliability of an FCA-regulated mortgage platform handling ~1.7M requests and 500K+ background jobs a day across 8 services. Ruby, Rails, PostgreSQL, Redis, Sidekiq, AWS, Bedrock, NewRelic, Sentry, Docker.

- Built and productionised a **unified case-management tool** that replaced the 22 separate systems mortgage brokers consulted per case with one centralised workflow, cutting case-handling time from **~65 minutes to ~10 minutes** so brokers can advise applicants live on a call — designed to double mortgage capacity without added headcount.
- Engineered **AI bank-statement analysis on AWS Bedrock** to extract income and risk signals from PDF statements for mortgage fact-finding, solving a token-limit failure that silently dropped transactions and contributing the Bedrock provider upstream to the **ActiveAgent** open-source project.
- Leading a **third-party integration** that automates lender application submission so brokers stop re-keying data by hand, owning it end to end from partner evaluation and mapping 1,100+ data fields through to the build.
- Established **platform observability from scratch** with six NewRelic dashboards, including the board-level OKR instrument the company steers by, surfacing systemic issues such as an integration at ~40% success generating ~17,000 dead background jobs a week.
- Diagnosed and recovered a **partner sync silently broken for a month**, re-architecting it to per-user fan-out with retry resilience and landing the first clean overnight run at 14,213 users.
- Set architecture direction through a full platform review, cut the **deployment pipeline from ~2 hours to 35 minutes**, and upgraded Ruby 3.2 to 3.4 ahead of EOL.

## Backend Senior Software Engineer, **Indeed**

2022 – 2025

Led backend engineering initiatives on a platform serving 1M+ workers across US/UK markets. Ruby, Rails, PostgreSQL, Redis, Kafka, OpenAI.

- Transformed the penalty appeals system over 6 months with cross-functional product teams, **reducing support tickets by 80%** and eliminating wrongful bans through system redesign and improved logic.
- Led rapid response to **critical incidents affecting 100K+ workers**, coordinating data-migration crisis recovery and preventing platform disruption.
- Collaborated on an **AI chat interface** using OpenAI GPT and custom few-shot learning, enabling natural-language task completion on job-posting tasks.
- **Mentored engineers** through pair programming, 1-on-1s, and team-wide knowledge sharing, and onboarded new hires across all levels.

## Senior Software Engineer, **Ministry of Justice**

2020 – 2022

Built government services for millions of UK citizens. Ruby, Rails, PostgreSQL, GOV.UK PaaS.

- Built a service to **apply to court about child arrangements**, validated through extensive user research, improving access to justice for families.
- Led development of a **service to check when to disclose cautions or convictions**, translating complex legal requirements into custom algorithms with policymakers and legal experts.
- **Transformed court payment processing** with an online system that eliminated weeks-long delays, removed manual cheque handling, and enabled instant fund allocation to court budgets.

## Senior Software Engineer, **Creditspring**

2018 – 2020

Joined a pre-launch fintech startup and architected scalable infrastructure, growing from 0 to 100K+ paying customers (500K+ registered users) while maintaining reliability.

- Architected **event-driven systems** using Rails, PostgreSQL, Redis, and Sidekiq, ensuring complete financial auditability and scalability for affordable borrowing products.
- Led the operations team to identify and implement automation, reducing manual processes to enable efficient scaling.
- Established monitoring and reliability practices maintaining **99.9%+ uptime** during rapid customer growth.

## Software Engineer, **Government Digital Service**

2016 – 2018

Developed core GOV.UK features for 60M+ monthly visitors. Ruby, Rails, Elasticsearch.

- Led development of GOV.UK **Step by Step pages**, simplifying complex government processes into clear user journeys for millions of citizens.
- Designed and implemented a **taxonomy system** for GOV.UK content organisation, helping content creators improve information retrieval across departments.
- Enhanced GOV.UK **search with Elasticsearch**, optimising performance and accuracy for millions of daily searches and phasing out legacy systems.

## EARLIER EXPERIENCE

Software Engineer roles at Unii, Sticky9, HouseTrip, and Think Orange. Further detail available on LinkedIn or on request.

## PROJECTS & PUBLICATIONS

**Modular Rails: Architecture for the Long Game** — technical book on building modular monolith applications with Rails engines as a principled alternative to microservices; well received in the Ruby on Rails community.

**Building Your Own Roguelike: A Practical Guide** — technical book on game development and event-driven architecture in Ruby, with an accompanying open-source game.